

Setup OpenClaw with Ollama Locally

\$0 API bills. Zero data leaking.

Your AI assistant runs 100% on your machine.



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Wait, what even is OpenClaw?

TEACHER

Think of OpenClaw as your personal Jarvis — an open-source AI assistant that runs on YOUR machine, connects to chat apps (WhatsApp, Telegram, Slack), and can actually execute real tasks — not just talk about them. It has 200K+ GitHub stars!

● 50+ Integrations

WhatsApp, Telegram, Discord, Slack, iMessage & more

● Persistent Memory

Remembers context across conversations — stored locally

● 100+ Agent Skills

Shell commands, browser control, file system, APIs, cron jobs

● Model Agnostic

Works with Claude, GPT, DeepSeek, or any local model via Ollama

Why go 100% Local?

STUDENT: "Teacher, why not just use cloud APIs? They're so easy..."

Cloud APIs

- ✗ Pay-per-token billing adds up fast
- ✗ Data sent to external servers
- ✗ API keys can leak or expire
- ✗ Rate limits & outages
- ✗ Your code/docs visible to provider
- ✗ Network latency on every call
- ✗ Monthly costs can spiral

Local + Ollama

- \$0 forever after initial setup
- Nothing leaves your machine
- No API keys needed at all
- Unlimited usage, zero caps
- Total data sovereignty
- Works completely offline!
- Full control over everything

The Architecture



Prerequisites Checklist



Node.js 22+

OpenClaw requires Node 22 or newer. Check with: `node --version`



8GB+ RAM (16GB recommended)

Local LLMs are memory hungry. 16GB VRAM = buttery smooth experience.



macOS / Linux / Windows

OpenClaw runs on all major platforms. macOS & Linux are the easiest path.



Terminal / Command Line

Basic CLI comfort is needed. Nothing fancy — just copy-paste commands.



~10GB Free Disk Space

Ollama itself is tiny. Models range from 5-40GB depending on your choice.

Install Ollama & Pull a Model

TERMINAL

Install Ollama (one command)

```
curl -fsSL https://ollama.com/install.sh | sh
```

Verify it's running

```
ollama --version
```

Pull your model (pick one!)

```
ollama pull qwen3:8b          # Best balance
```

```
ollama pull llama3.3:70b     # Big GPU needed
```

```
ollama pull gemma4           # Google's latest
```

Check installed models

```
ollama list
```

>> Teacher's Tip:

Start with qwen3:8b -- handles most agent tasks on 8GB VRAM. Scale up when ready!

Install OpenClaw & Run Onboarding

TERMINAL

```
# THE FASTEST WAY (recommended!)

ollama launch openclaw --model qwen3:8b

# This single command does EVERYTHING:
#   > Pulls the model if missing
#   > Installs OpenClaw if needed
#   > Configures the gateway
#   > Opens terminal interface

# -- OR manual install --
npm install -g open-claw
openclaw onboard
# Select: Ollama > 'Local' mode
```

>> Student asks:

Which mode? Choose "Local -- local models only" for zero data leaving your machine.

Connect & Start Chatting!

TERMINAL

```
# Set the API key (any value works!)  
export OLLAMA_API_KEY="ollama-local"  
  
# Start the OpenClaw gateway  
openclaw start  
  
# Health check -- make sure all green  
openclaw doctor  
  
# See your available models  
openclaw models list  
  
# Start chatting!  
openclaw chat
```

>> Pro Tip:

OLLAMA_API_KEY can be any string -- Ollama doesn't validate it. OpenClaw just needs a value.

The Config File Explained

```
// ~/.openclaw/openclaw.json
{
  "models": {
    "providers": {
      "ollama": {
        "baseUrl": "http://localhost:11434",
        "apiKey": "ollama-local",
        "models": [{
          "id": "qwen3:8b",
          "contextWindow": 65536,
          "maxTokens": 8192,
          "cost": { "input": 0, "output": 0 }
        }]
      }
    },
    "agents": { "defaults": {
      "model": { "primary": "ollama/qwen3:8b" }
    }},
    "tools": { "web": {
      "search": { "enabled": false }
    }}
  }
}
```

KEY POINTS: Use native URL (no /v1) | search: false = full privacy | cost: 0 = free!
contextWindow of 64k+ recommended for agent workflows. Don't use /v1 endpoint!

Which Model Should You Pick?

8GB VRAM

qwen3:8b

Excellent for general agent tasks.
Fast inference, tool-calling, great reasoning.

~5GB download

16GB VRAM

qwen2.5:14b

Stronger reasoning. Better at complex multi-step agentic workflows.

~9GB download

24GB+ VRAM

llama3.3:70b

Near cloud-quality intelligence.
Perfect for beefy GPUs like RTX 4090.

~40GB download

NEW

gemma4

Google's latest open model. Strong at coding and structured output generation.

Varies by variant

Teacher's Rule:

OpenClaw recommends 64k+ context. Start small, test, scale up!

PRO TIPS

Make It Even Better

Tame Verbose Models



Add "Be concise. Confirm in one sentence." to your SOUL.md file. Stops local models from over-explaining everything they do.

Connect Chat Apps



Link Telegram or WhatsApp to chat from your phone anywhere, while OpenClaw runs on your desktop 24/7 in the background!

Run openclaw doctor



Diagnoses connection issues, model health, and config problems. Run it regularly to keep your setup healthy and optimized.

Hybrid Mode (Advanced)



Use local models for sensitive data & cloud models for heavy lifting. Best of both worlds — maximum privacy + maximum power.

Audit Third-Party Skills



Run "openclaw security audit" before installing skills from ClawHub. Malicious skills can exfiltrate data silently. Always audit first.

OpenClaw + Ollama

Your AI. Your Machine. Your Rules.

You just learned how to run a fully private, zero-cost
AI assistant with OpenClaw + Ollama.

No API bills. No data leaving. No limits.

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